



Time series signal forecasting using artificial neural networks: An application on ECG signal

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ABSTRACT

Time Series Forecasting is the prediction of future values of a signal based on the observed past values. It has various applications in signal processing, especially in the medical field which needs high accuracy. This paper presents an MLP (Multilayer Perceptron), a class of FFNN (Feedforward Neural Network) for highly accurate time series forecasting. There are various methods of signal processing that are used in time series forecasting but each method is specific to the particular problem it solves. The current methods involve the use of different types of adaptive filters out of which the most common method is LMS (Least Mean Square) algorithm. Although the adaptive filters give a decent accuracy, but neural networks (NN) give the results more than satisfactory. On performing time series forecasting on a simulated ECG (Electrocardiogram) signal, an accuracy of 95.72% was achieved using ANNs (Artificial Neural Networks) competing with the LMS filter, which gave only 79% accuracy. When the same was implemented on real ECG data of a person suffering from Sleep Apnea, the ANNs offered 98.68% while LMS filter displayed only 91% accuracy. Additionally, the neural network was also denoising the signal while predicting. A signal-to-noise ratio of 29.71 dB and 16.33 dB for Neural Network prediction and LMS filter prediction was attained, respectively. In the case of the real data, the aforementioned values stand at 22.8 dB and 3.8 dB, respectively. Simulated results show that the neural networks give superior performance in time series forecasting than Adaptive Filters.

1. Introduction

A 'Time series' is a series of data points that occur in sequential order over a period, whereas 'Forecasting' is a prediction of the future. To explain it in simple terms it can be said that Time Series Forecasting is the prediction of future values of a series based on its past values. ECG signal is a time series data. It is measured in equal intervals of time at various frequencies. An ECG is the electrical signals generated by a human heart as it beats. It is a crucial time series data in the medical field. It reveals critical details about a patient's health which may help diagnose or monitor sicknesses.

Applications of time series forecasting range from signal processing to statistical analysis. To name a few, it could be predicting the weather pattern, crop yield, stock prices, birth rate, etc. [1–5]. It can play a pivotal role in biomedical practices. An example of that could be the prediction and monitoring of ECG. It is important that the ECG is always in rhythm and maintains a proper pace. An ECG can detect cardiomyopathy, heart attacks, coronary heart disease, arrhythmias [31], high blood pressure, previous history of heart attacks, etc. Hence, proper

monitoring techniques are cardinal. In the current work, an ECG signal is predicted with two different approaches to compare which technique gives the maximum accuracy. As a matter of fact, if high accuracy is achievable in any applications of time series forecasting, then the same predictions can help in taking precautions or for making the right decisions at the right time. Hence, the ease of forecasting and high accuracy will always remain of paramount importance.

Adaptive filters being at the heart of signal processing offer numerous methods to perform signal forecasting. An adaptive filter is a system that includes a linear filter with a transfer function that is controlled by variable parameters and has the ability to alter those parameters using optimization techniques. For example, Least Mean Squares (LMS) Based FIR (Finite Impulse Response) Adaptive Filters, Recursive Least Squares (RLS) Based FIR Adaptive Filters, Affine Projection (AP) FIR Adaptive Filters, FIR Adaptive Filters in the Frequency Domain (FD), Lattice-Based (L) FIR Adaptive Filters are a few adaptive filters. Apart from adaptive filters, ML (Machine Learning) can also be used to predict signals. One such ML technique is discussed in this paper.

a. Artificial neural networks (ANN)

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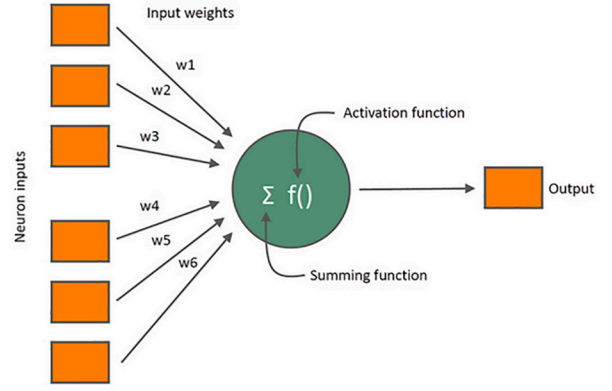
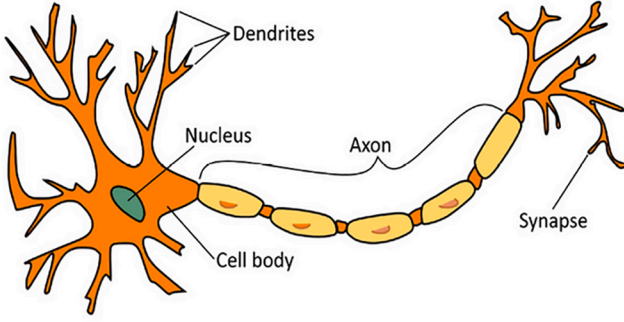


Fig. 1. Biological neuron vs artificial neuron.

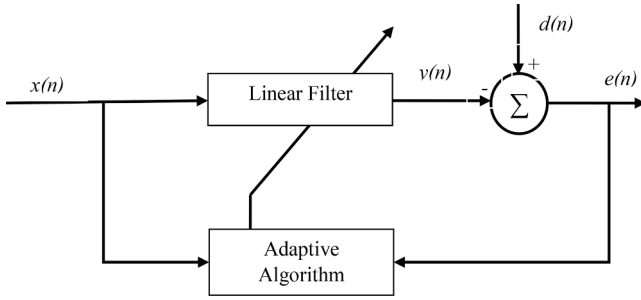


Fig. 2. Adaptive filter.

The term “Artificial Neural Network” is derived from biological neural networks that compose a human brain as shown in Fig. 1. They were created to mimic the neural networks which make up the human brain, hoping to allow the machines to make decisions like humans. Just like our brains, it may have multiple layers of networks. It has the below-listed layers:

- 1) Input Layer
- 2) Hidden Layers
- 3) Output Layer

The input goes through a series of transformations in the hidden layer, which finally results in output that is conveyed through the output layer. The artificial neural network takes input vector $X = [x_1, x_2, x_3 \dots x_n]$ and computes the weighted sum of the inputs and adds a bias ‘b’ to it where weights are $W = [w_1, w_2, w_3 \dots w_n]$ [3]. This computation can be represented in Eq. (1).

$$z = \sum_{i=1}^n w_i * x_i + b \quad (1)$$

The computed value ‘z’ is fed into the activation function as an input to give out the final output. The activation function decides whether a given neuron must be fired or not.

$$y = f(z) \quad (2)$$

↓
Activation function

A representation of the activation function is given in Eq. (2) where f is any activation function and y is the final output.

b. Adaptive filters

Digital filters whose coefficients change to achieve an optimal state are called Adaptive Filters. A cost function is used as a criterion to optimize. A block diagram for an adaptive filter is shown in Fig. 2. There are many types of cost functions but almost all of them need the error value to be computed. The most common adaptive filter is the LMS filter

[6–8].

LMS filters are a category of adaptive filters that use gradient descent to narrow down the mean squared error. It’s filter coefficients adapt based on the instantaneous error and the rate at which they update is determined by the learning rate. They are made up of an ADALINE (Adaptive Linear) network.

Knowing the future can always allow us to be better prepared. Specifically, while forecasting signals, the conventional methods that use adaptive filters are not very satisfactory in terms of accuracy. Although they give an appreciable accuracy, they are far from desired value needed in critical applications such as in the biomedical field. Hence, this calls for a need to come up with a better method for time series forecasting [2,5].

The main contributions of the present study are as follows:

- This study aims to perform time-series prediction using neural networks, contrary to the traditional method of using adaptive filters offered by signal processing; and thereby obtain better and more accurate results.
- This was achieved by employing time-series forecasting using neural networks on simulated as well as on real time ECG Signal of sleep apnea patient and comparing them with the traditional methods.

The rest of the paper is structured as follows: Section 2 presents a review of the literature and related work of time series forecasting. Section 3 describes the proposed method followed by Section 4, which gives the obtained results, and a discussion of the results and a practical use case scenario is described. Finally, the conclusion and future scope are drawn in Section 5.

2. Related works

Dipankar Gupta [2] in a thesis mentions about time series analysis of ECG data and its unreliable forecasting. According to the study, linear systems that are used to describe complex biological systems such as ECG data are no longer satisfactory. It was mentioned in the paper that “It is assumed that parameters of our model are known exactly but in practice parameters of the model are estimated. So, some of the error made in forecasting will be due to the error of the estimation of the parameters rather than just to the random error of the model.”

In Eva Volna et al. [26], the implementation of classification of ECG is discussed. The state of the patient is diagnosed as to whether sick or healthy, by classifying the ECG into 0 or 1. The concept of the paper was to detect cardiac abnormality based on the pattern recognised in the ECG i.e. to detect normal cycle or arrhythmia and make further diagnosis based on it. The classification was performed in three different methods, and one of them was to use Neural Networks.

In accordance with [27], it can be said that deep neural networks (DNNs) prove to be good for analysing the ECG signal where it

outperforms the cardiology resident medical doctors in diagnosing 6 diseases. The paper suggests eradicating the two-step approach to make any diagnosis using ECG where, first the features are extracted from the ECG using conventional techniques and later used for classifications. The NNs perform “end-to-end” learning where it independently extracts the features of ECG and learns the nature of it [19].

In Sireesha et al. [7], adaptive filter based on Least Mean Square (LMS) algorithm to be applied in underwater acoustic communication. The paper discusses the system configuration, filter structure, and the implementation of the Adaptive LMS algorithm. The factors such as the convergence and stability of the filter which are responsible for stable adaptation behaviour are also discussed. The performance of the designed adaptive filter is compared with the in-built MATLAB LMS filter. The study was found to be satisfactory and viable.

Zeidler [9] mentions that the conditions required to implement real-time adaptive prediction filters that provide nearly optimal performance in realistic input conditions are delineated. The paper explicitly studies the effects of signal bandwidth, input signal-to-noise ratio (SNR), noise correlation, and noise non-stationarity. It also shows that the signal processing gain nonlinearly degrades, as a function of input. It mentions that the stochastic nature of the weights bound the performance of the adaptive filters in the optimal filter state.

In Oancea and Ciucu [10], it has been demonstrated that a NN can approximate any continuous function. Forecasting of financial data series is successfully implemented with Neural networks. It states that the classical methods used for time series prediction like Box-Jenkins or ARIMA (Autoregressive Integrated Moving Average) assumes that there is a linear relationship between inputs and outputs. Neural Networks are said to have the advantage that can approximate nonlinear functions. In this paper they compared the performances of different feed forward and recurrent neural networks and training algorithms for predicting the exchange rate EUR/RON and USD/RON.

To discuss a few research works where the adaptive filters were used for prediction, it can be seen in the paper [29], that Adaptive filtering to predict the lung tumour motion during breathing was performed using linear adaptive filters as well as adaptive neural networks algorithm. Wesen et al. [28], had designed an adaptive filter for stock market prediction using a correlation-based criterion. The research proposes novel adaptive filtering-based predictors for investment strategies to be used in the stock market as a decision-making engineering tool. In Shavelis et al. [30], the prediction of the future values of the vehicle sideslip angle is performed using neural networks and normalised least mean squared algorithm and compared their performance.

In Putra and Kosala [11], artificial neural networks were found to be delivering better forecasting performance than the results obtained by the well-known ARIMA technique. They believe that, unlike conventional techniques for time series analysis, an artificial neural network

needs little information about the time series data and can be applied to a broad range of problems. Artificial Neural Networks are suitable for many tasks in pattern recognition and ML [12,13]. In this paper, an APL (Automated Predictive Library) system for forecasting univariate time series with artificial neural networks is presented.

In Xiao et al. [39], discussed how to use of different ML models could be used to predict UCG (Underground Coal Gasification) to contribute to clean and efficient use of coal and its stable production. Dual-source long short-term memory (LSTM) model was proposed as the optimal solution for the application. In Xiao et al. [40] it is described how the insufficient time dependency in MTS (Multivariate Time Series) could be tackled by the use of a combination of attention mechanism and LSTM while performing MTS prediction.

In Kaushik et al. [14], the primary objective was to evaluate different statistical, neural, and ensemble techniques in their ability to predict patients’ weekly average expenditures on certain pain medications. Two statistical models, persistence (baseline) and autoregressive integrated moving average (ARIMA), a multilayer perceptron (MLP) model, a long short-term memory (LSTM) model, and an ensemble model combining predictions of the ARIMA, MLP, and LSTM models were calibrated to predict the expenditures on two different pain medications. Results revealed that the ensemble model outperformed the persistence, ARIMA, MLP, and LSTM models across both pain medications [15,16].

Carrying out analysis of ECG has been ever prevalent. As medical fields always demand precision, efforts remain to continuously improve the technology to make lives of humans last longer. In Zhao et al. [32] the implementation of noise rejection for wearable ECGs using a combination of MFSW (Modified Frequency Slice Wavelet Transform) and CNNs (Convolutional Neural Networks) was discussed. A classification model was used to detect the noisy segments of the signal. The paper [33], displays a comprehensive and experimental study on classification of ventricular premature and ischemic heartbeats with the help of various methods including the use of ML models. In a different article [34], observations were recorded as cardiac arrhythmia was detected from ECG using adaptive feature extraction and modified Support Vector Machines (SVMs). There are many more such papers published to improve the sense making process from the biomedical signals [35,36]. Some papers such as [2,9,18,37] even though old, prove to be a good source of resources of knowledge to bridge the gap between what was once known and what is being studied.

There have been quite a few theses and papers published on time series forecasting. Most of them were performed on statistical data using statistical methods such as ARIMA model (Auto-Regressive Integrated Moving Average) [18] but to the best of our knowledge, time series signal prediction has not been reported in literature. Considering ANNs, even it has a lot of papers published on its technology but most of them compared the performance of several techniques present internally in the ANN technology [39]. Moreover, even adaptive filters have had their share of papers published, but even here the focus was mostly laid on comparison of different types of adaptive filters. But our solution comes from a field that is a union of all three previously mentioned areas, i.e. time series forecasting of a signal using ANNs and adaptive filters [17]. In our proposed solution we perform time series forecasting of an ECG signal using ANNs, as well as with conventional adaptive filters. Further, performance is compared. Proper implementation of the method may even eliminate the necessity of signal de-noising before analysis.

3. Proposed technique

The proposed solution involves forecasting a time series using artificial neural networks, to be more specific MLP (Multilayer Perceptron), a class of FFNN (Feedforward Neural Network) is used to achieve high accuracy. Although the adaptive filters provide a decent accuracy, the neural networks give us a more than satisfactory result proving to be better than the adaptive filters.

Forecasting of a simulated ECG signal is performed using ANNs as

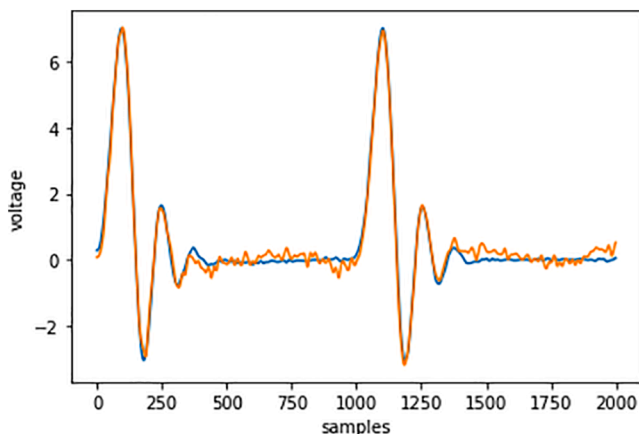


Fig. 3. ECG visualization.

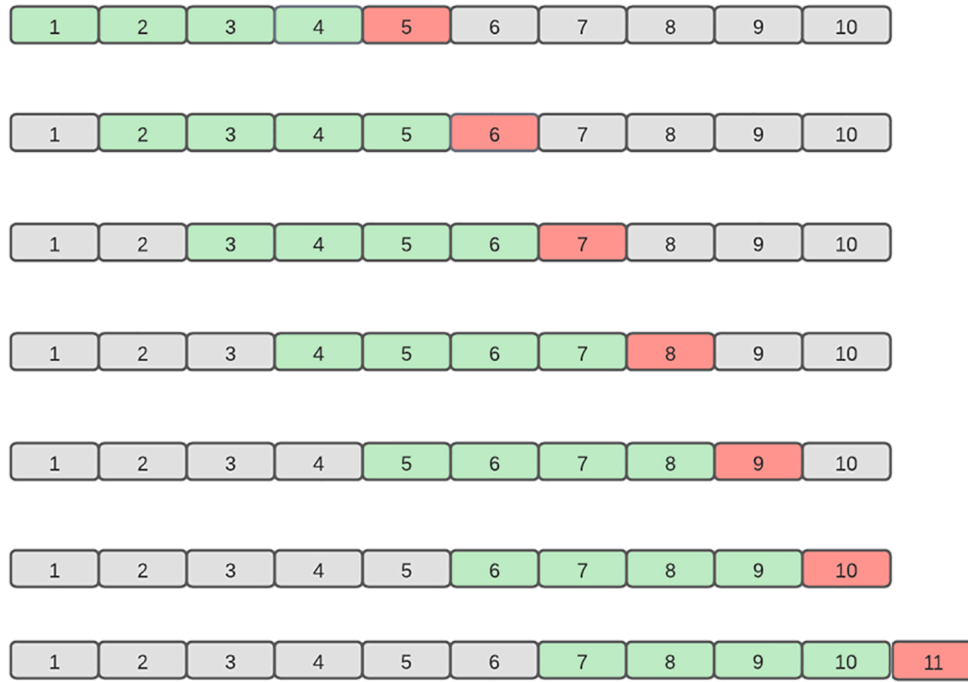


Fig. 4. Sliding window technique.

well as LMS filters. Further, their performances are compared. Our proposed solution along with giving highly accurate forecasting, also denoises the signal [20]. Further, the same is also applied to a real ECG data of a sleep apnea patient and its outcomes are discussed in the ‘results and discussion’ section.

a. Data Pre-processing

i. Data Generation

In the present study, simulated data has been used. A library called ‘Neurokit2’ was implemented to simulate the required ECG data [21]. The library is a user-friendly package providing easy access to advanced bio-signal processing routines. It also allows users to add noise by a user-defined factor. A noise introduced simulated ECG signal is used to predict the upcoming sample value of the noiseless ECG data.

ii. Visualization of ECG

In Fig. 3, the x-axis represents the samples and y-axis represents the Voltage. Yellow signal is the noisy signal whereas the blue signal is the original signal.

Note: As it is a simulated ECG signal it may not be exactly like an original ECG. The implementation presented in this paper is just a proof of concept for the methodologies suggested. However, the proposed method is also implemented and discussed on real ECG data of a sleep apnea patient in the ‘results and discussion’ section.

iii. Train and Test Split

To evaluate the model, the available data is split into two parts. The first subset is used to fit the model and is called as training dataset. Whereas the second subset is not used to train the model; instead, it is used to later test the model by making inferences using data never seen by the model [22]. This surprise test for the model reveals the true accuracy of the model. Hence, the second subset is called test data.

Usually, 75% of data is used for training and 25% of data is used for testing. Hence, the 20,000 data points are split into 2 subsets of 15,000 and 5000 for training and testing, respectively.

iv. Sliding Window Technique

As its name suggests, this technique involves taking a subset of data from a given array or string, expanding it and adding in a new value to the subset, and removing one beginning value, hence causing the sliding effect. For example, consider a list of numbers from 1 to 10.

A window of size 4 is taken. So, the sample values in the first window

of size 4 are 1,2,3,4 as shown in the Fig. 4. The green block represents the current windows, and the red block is the predicted values. Now after sliding the window to the right by one number, which is nothing but removing the first number from the previous window and adding a new number on its right a new window of size 4 whose values are 2,3,4,5 is obtained. And for every 4 values the 5th value is predicted. In this way, 7 batches of size 4 each can be formed. Which would result in $7 \times 4 = 28$ points of data, which is far greater than 10. Meanwhile, it also allows us to predict 7 consecutive values.

In the presented study, a window size of 64 was used. The generated time series data only consisted of 15,000 samples but through sliding window technique, $14936 \times 64 = 955904$ samples of input data were created for the neural network so that the model had enough values to learn from.

b. Time Series Forecasting using Neural Networks

i. Model Specifications

The proposed model uses a 3-layer deep sequential feed forward neural network. The first layer consists of 32 neurons, 64 samples are fed as input to the first layer where they are mapped to 32 samples. Now, the 32 samples produced by the first layer are fed into second /hidden layer which produce 8 samples. Further, the 8 samples are fed into the last /output layer, giving out a single output.

Going into further specifications, the model uses a ReLu Activation Function. The following points explain the purpose of an activation function:

1. It decides whether a neuron must be fired, which gives the ANNs a decision-making ability.
2. It adds non-linearity to the network, allowing the ANNs to understand complex patterns.
3. It helps in restricting the output of a neuron to avoid computational issues.

The chosen activation function, ReLU stands for Rectified Linear Unit. It outputs zero if the input is negative and acts as an identity function if the input is positive. The equation for the ReLU function is given by Eq. (3) where x is the input given to the function.

$$f(x) = \max(0, x) \quad (3)$$

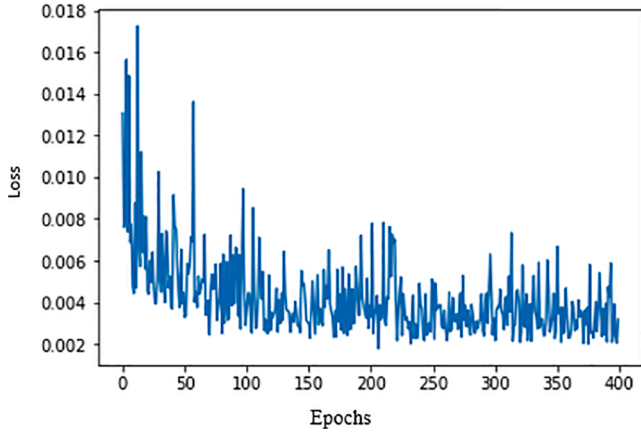


Fig. 5. Loss vs Epochs graph.

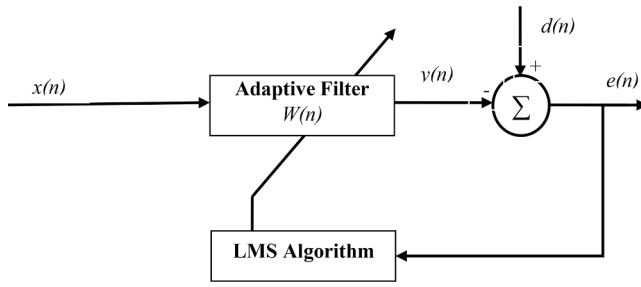


Fig. 6. LMS filter block.

Adam optimizer is used for tuning the weights and biases in order to achieve optimal output. It is a stochastic gradient descent algorithm that is based on adaptive estimation of first and second order moments. The loss function used is Mean Squared Error. Mean squared error (MSE) is the most used loss function for regressions. The loss is the mean overseen data of the squared differences between true and predicted values or writing it as a formula. If $Y = [y_1, y_2, y_3, \dots, y_n]$ is a vector of true values and $Y^p = [y_1^p, y_2^p, y_3^p, \dots, y_n^p]$ is the vector of the predicted values, then the means squared error is given by Eq. (4), where y is the i^{th} true value and y^p is the i^{th} predicted value.

$$MSE = \frac{\sum_{i=1}^n (y_i - y_i^p)^2}{n} \quad (4)$$

The training data is fed into the model. The training is implemented using the *fit()* method. The layers implement the activation function that

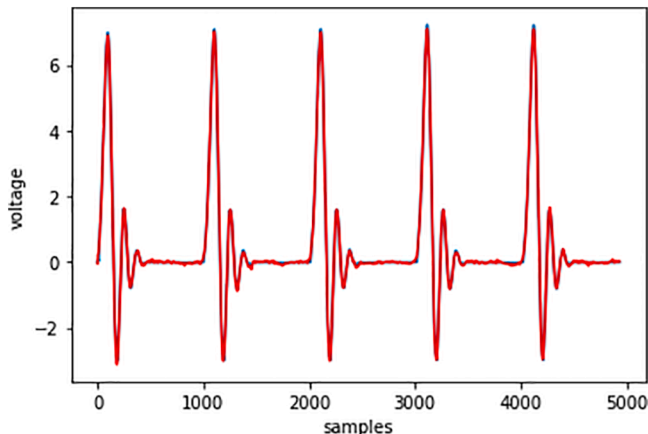


Fig. 7. Prediction vs Original Signal graph (using ANNs).

was mentioned earlier. Optimizer tunes the weights and biases. The model is trained until it does not show any further significant improvements.

In Fig. 5, the x-axis represents the epochs, and the y-axis represents loss. It can be seen that the loss is reducing as number of epochs progress.

After training, the model needs to be tested. To test the model the *predict()* function is run on the test data and the predictions are made. These values are further used to calculate the performance of the model i.e., accuracy.

c. Time Series Forecasting Using LMS Adaptive Filter

Least mean square filter is an adaptive filter where the coefficients are updated in accordance with the means squared error such that it is least, as shown in Fig. 6.

The predicted, i.e. the future value $y(n)$ is the dot product of previous values $x(n)$ and its corresponding coefficients vector $W(n) = [w_1, w_2, w_3, \dots, w_n]$ where ' n ' is the instantaneous time. The error is then calculated, and the filter coefficients are updated accordingly. The output sample will be as given in Eq. (5).

$$y(n) = W(n) \cdot x(n) = \sum_{i=1}^n w_i \cdot x_i \quad (5)$$

An error ' e ' is generated to represent the deviation of the output from the desired value, as in Eq. (6) where ' d ' is the desired value at instantaneous time ' n '.

$$e(n) = d(n) - y(n) \quad (6)$$

This error signal is fed back to the network to update the weights to effectively predict the future samples. Now, the LMS filter makes use of the LMS algorithm to update the weights. It is based on the instantaneous values of the cost function given by Eq. (7), where C , e are cost function and error at instantaneous time ' n ', respectively.

$$C(n) = \frac{1}{2} (e^2(n)) \quad (7)$$

Now, to understand the change in cost function with respect to change in the weights the following computations are done.

$$\partial C(n) / \partial w = e(n) \cdot \partial e(n) / \partial w$$

$$\partial e(n) / \partial w = x(n) \text{ (from Eq. (6) and Eq. (5)).}$$

$$\partial C(n) / \partial w = -x(n)e(n) \quad (8)$$

Eq. (8) is called as gradient approximation. Now, the updated weight is given by Eq. (9) where η is the adaptation constant that determines the convergence rate [25].

$$W(n+1) = W(n) + \eta C(n)$$

$$W(n+1) = W(n) + \eta X(n)e(n) \quad (9)$$

The above working is implemented, and the required metrics are computed and noted.

4. Results and discussion

The accuracy metric used is Mean Absolute Error (MAE). In time series analysis, the mean absolute error is a popular metric for forecast error. In this paper, the mean of the absolute error is first calculated, and then for ease of understanding the accuracy is calculated in percentage. The MAE initially gives us the extent of mistakes made in the predictions. It is a value between [0,1]. 0 means that there were 0 errors made and 1 depicts that all the predictions were wrong. To calculate MAE, the number of falsely predicted values is divided by the total number of predictions ' n '. Its mathematical representation is given in Eq. (10) where $Y_{predicted}$ represents the predicted values and Y_{true} represents the true values.

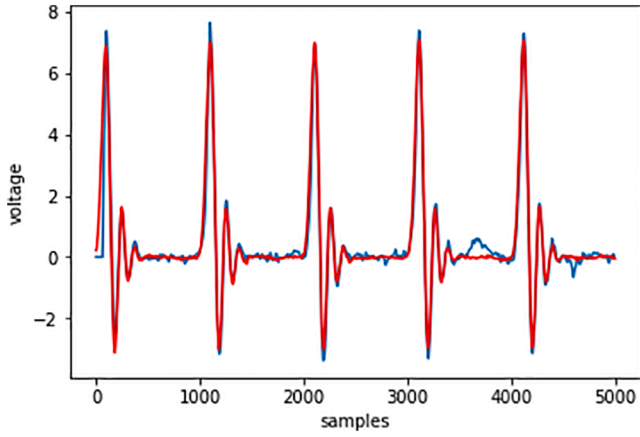


Fig. 8. Prediction vs Original Signal graph (using LMS filter).

$$MAE = \sum_{i=1}^n \frac{|Y_{predicted} - Y_{true}|}{n} \quad (10)$$

After calculating the MAE, the obtained value is then subtracted from 1 and then multiplied by 100 to convert it into percentage as given in Eq. (11).

$$Accuracy (\%) = (1 - MAE) * 100 \% \quad (11)$$

a. Accuracy of ANNs

After a lot of training and optimization the ML model accomplishes an accuracy of 95.5%. The accuracy metric used is Mean Absolute Error. Accuracy metrics is the proportion of the total number of predictions that were correct. Mean Absolute Error as its name suggests is the mean of the absolute error, it identifies how big of an error could be made.

In Fig. 7, the x-axis represents the samples, and the y-axis represents the Voltage. The red graph is of the original signal and the blue graph is the forecasted signal. Although it seems to be perfectly overlapping, there are little peaks of blue in the graph. From the graph, it is clear that the Neural Networks are performing exceptionally well.

b. Accuracy of LMS Filter

An accuracy of 79% is achieved using the adaptive filter. The same accuracy metric is used as in the case of ANNs so that the comparison is fair.

In Fig. 8 the x-axis represents the samples, and the y-axis represents the Voltage. The red line in the graph is of the original signal and the blue line in the graph is the forecasted signal. Although it seems to be doing a decent job, it does not look satiable [18]. Especially in the case

of biomedical applications, there is no place for far from perfect accuracy.

c. Comparison of LMS Filter and ANNs

Visually comparing the Figs. 7 and 8, it can be seen that although LMS filter is doing a decent job, ANNs overlap almost perfectly with the original signal.

Getting into the numbers, Accuracy score in this case is nothing but the mean absolute error of the selected number of samples. An ideal value for it would be 0. The accuracy scores stand at 0.04 and 0.2 for ANNs and LMS filter prediction, respectively. Clearly, ANNs are closer to the ideal value [5]. From the accuracy scores, accuracy percentages are further calculated. An accuracy percentage of 95.53% is achieved for the forecasting done by the ANNs and an accuracy percentage of 79% is obtained for the forecasting done by the LMS filter [14].

Looking at the numbers accomplished, it can surely be agreed that the neural networks give a better result than the adaptive filters. The least mean square error is considered to be a single neuron system as its working is similar to the way a neural network works. In spite of that, the neural networks give a way better result due to the provision of flexibility in choosing the number of neurons, number of layers, amount of data, optimization algorithm, loss metrics, etc.

Further, on plotting a graph to depict errors made per sample for both the methods, it can be observed from Fig. 9 that the ANNs produced very little deviation from the expected results in contrast with LMS filter.

As discussed earlier, the signal predicts the noiseless ECG signal based on the noisy ECG data. Hence, signal to noise ratio (SNR) was also computed as shown in Eq. (12). Neural networks method achieves an SNR of 29.35 dB while the LMS filter method gets an SNR of 16 dB. Clearly, the neural network performs better. To recall, neural network was trained using the noisy data, yet it predicted the noiseless data with a good accuracy and also gave a decent SNR ratio. Hence, it can be seen that the neural network along with predicting the future values, also de-noises the signal.

$$SNR = 10 \times \log \left(\frac{\text{Signal Power}}{\text{Noise Power}} \right) \quad (12)$$

Although it adds so much value in terms of accuracy and noise reduction, there are a few traits that are compromised [20,22]. It is an exceedingly difficult task to train using neural networks as there are so many ways in which they can be implemented. Proper architecture must be first decided as per the application and then it should be improvised further while observing the model's performance. Also, sometimes the model needs to be trained for exceptionally long time to give satisfactory results.

Moreover, if the amount of data being dealt with increases or if the

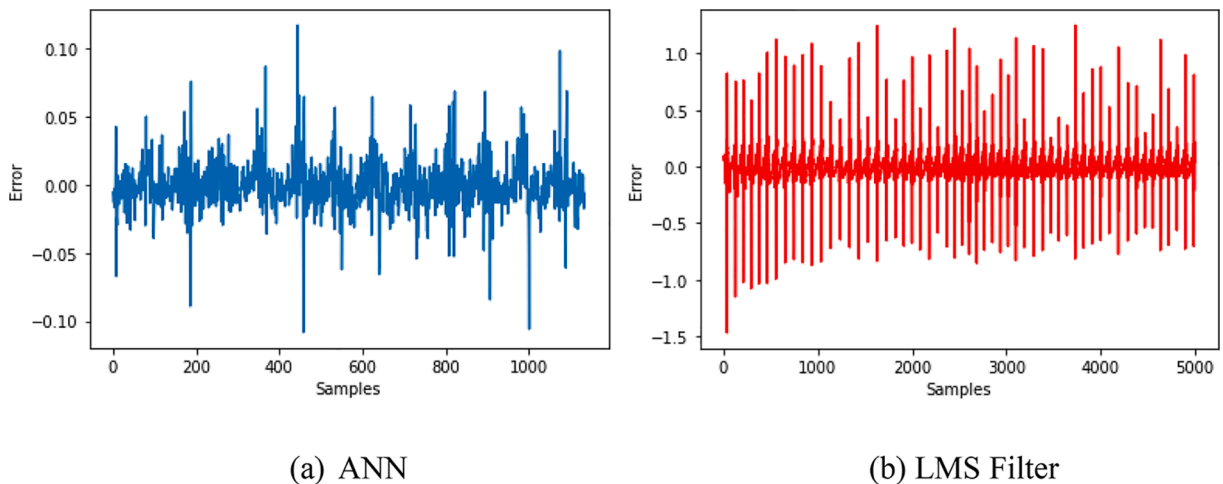


Fig. 9. Error vs sample graph.

Table 1

Comparison table between ANNs and LMS filter.

Parameter	TSF using ANNs	TSF using LMS filter
Accuracy	95.72%	79%
Noise Reduction	Good	Poor
SNR	29.71 dB	16.33 dB
Training	Difficult	Easier to implement.
Time Complexity	High	Low

number of layers in the neural networks increases, the time complexity becomes very high as it would increase the number of computations required [41]. Consider there are ‘ n ’ inputs followed by 3 layers of neural networks with the number of nodes in each layer as ‘ m ’, ‘ p ’, ‘ q ’ respectively. Now, the weights of the layers can be represented as W_{nm} , W_{pm} , W_{qp} . If ‘ b ’ is the number of training examples, then the first layer will need the following computations as given in Eq.13 (ignoring addition of bias for the ease of computation and understanding) where U_{mb} is the output of the first layer before applying activation function while, Z_{nb} is the input vector. In the current study ‘ b ’ is given by Eq.14 where N is the length of signal and ‘ n ’ is the window size.

$$U_{mb} = W_{nm} \times Z_{nb} \quad (13)$$

$$b = (N/n) \quad (14)$$

The above operation would have a time complexity of $O(m*n*b)$, as it is a matrix multiplication. Then the activation function is applied to produce the inputs to the next layer as below.

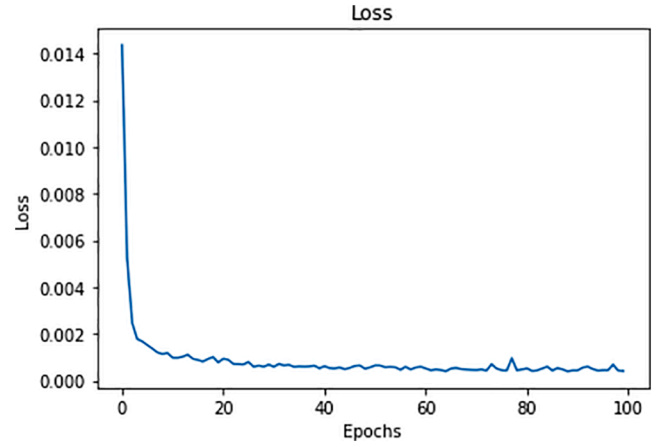
$$Z_{mb} = f(U_{mb}) \quad (15)$$

This operation has a complexity of $O(m*b)$ as the operation is done on every element in the matrix. Overall, we would get $O(m*b*n)$ if we simplify it as mentioned in Eq. (16).

$$[O((m*n*b) + (m*b))] = O(m*b*(n+1)) = O(m*b*n) \quad (16)$$

Similarly, for the consecutive layers, the time complexity would be $O(m*p*b)$ and $O(p*q*b)$. The final complexity would be a sum of all the intermediate complexity multiplied by the number of epochs, as the whole process repeats for every epoch. If e is the no. of epochs then, time complexity would be $O(eb*(nm + mp + pq))$.

With the current study as the context, the neural network had to make 2312 multiplications, 41 additions, and application of activation function another 41 times in each epoch. Training an ML model may take several epochs before converging at a considerable accuracy on top of that hyperparameter tuning is also necessary. Meanwhile, for the

**Fig. 11.** Training loss.

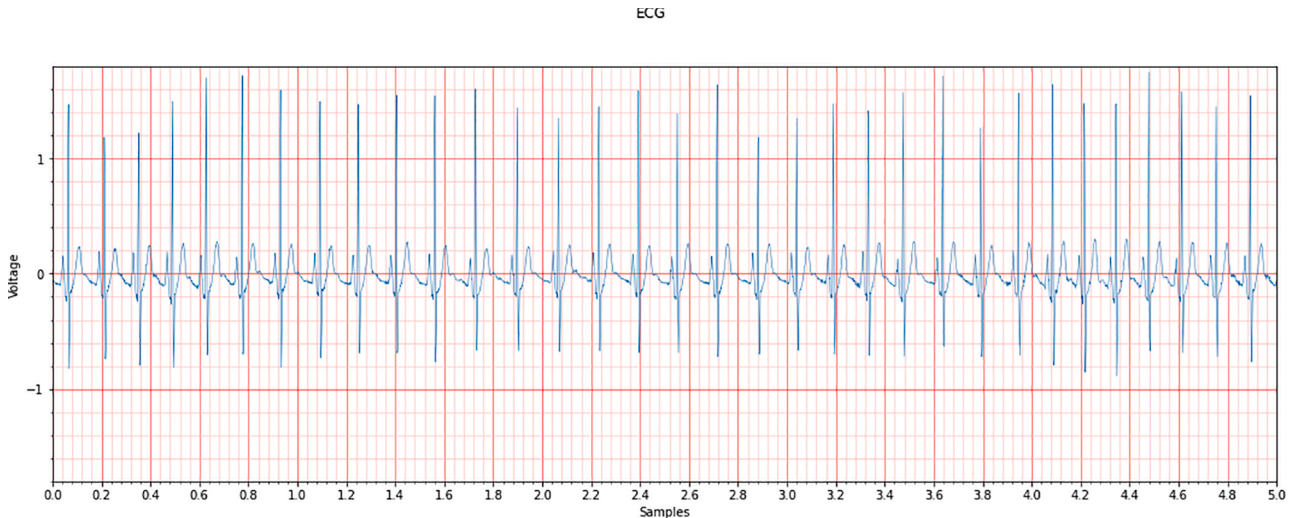
adaptive filters, the time-consuming tasks could be finalising the algorithm to use and implementing the algorithm. Although, even adaptive filters require some computations and parameter tuning, they are much less as compared to the neural networks. The LMS filter has a constant time complexity of $O(N)$ as discussed in [38]. Since, we are making a prediction for every window, the complexity of LMS filter specific to the study's use case would be $O(N*n)$. Hence, keeping all this in mind, it can be said that ANNs demand more time and attention than the Adaptive filters. But with continuously evolving hardware technology, time complexity can be improved with the help of various accelerators.

From the above discussion, the following differences between time series forecasting using ANNs and LMS filter can be tabulated as shown in Table 1.

d. Practical Use Case Scenario

To verify scope of the current research work in a practical scenario, the proposed solution was implemented on the ECG of a person suffering from sleep apnea. Sleep Apnea is a sleeping disorder where the patient experiences halt in their breathing during sleep which often leaves them tired even after the nap. If it is not treated, it may lead to obesity, hypertension, heart attack, heart failure, etc. As the person stops breathing, the heart rate starts to decrease, or the person may even develop irregular heartbeat. Hence, ECG analysis can assist diagnose sleep apnea and determine the causes of cardiac arrhythmias [23].

The ECG data was acquired from <https://archive.physionet.org/cgi-bin/atm/ATM>, where healthcare professionals and many others

**Fig. 10.** ECG of a sleep apnea patient.

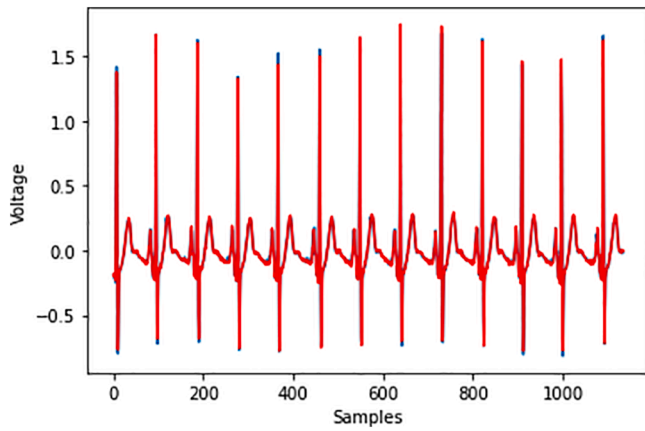


Fig. 12. Prediction vs Original Signal graph (using ANNs).

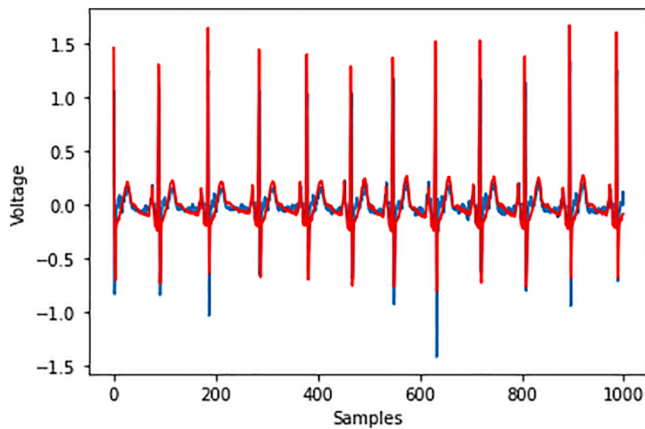


Fig. 13. Prediction vs Original Signal graph (using LMS filter).

upload data. It acts a data hub for many researchers. The data was recorded for around 1 min with as a frequency of 100 Hz. Fig. 10 shows the ECG sample values plotted against their amplitude.

For predicting with Artificial Neural Networks, the same architecture used for prediction on simulated ECG was used. Fig. 11 depicts the drop in the loss as the epochs increase.

Even in this case, the ANNs displayed a superior performance giving an accuracy of 98.68%. The performance of the ANNs is plotted in the Fig. 12, the x-axis represents the samples and y-axis represents their amplitude.

On predictions done using LMS filter, an accuracy of 91% was achieved. The performance of LMS filter is plotted in the Fig. 13. The x-axis represents the samples and y-axis represents their amplitude.

Considering the SNR, the ANNs give 22.8 dB SNR, while LMS filters

only give 3.8 dB. It is evident from the below graphs shown in Fig. 14 that ANNs make less errors than the LMS filters proving to be more reliable for forecasting.

e. Architecture Required

To develop an ECG prediction system in real time there are four main layers to be implemented as shown in the Fig. 15 [24]. The first one is the data acquisition layer which may use different kinds of sensors to acquire ECG data like electrodes, wearable devices, etc. Then second layer, i.e., the pre-processing layer must be carried out. In this layer the data is pre-processed, the required noise removal, transformation or compression of data are performed. Then comes the modelling and analytics layer, where efforts are made to extract meaning from the data. Finally comes the Visualization Applications and Interface layer where the findings of the previous layer are delivered to the users. The users may be individuals or hospitals. The computing technology may differ based on the context.

5. Conclusion and future scope

The current work was aimed at finding a method to achieve highly accurate time series forecasting of signals with the help of neural networks and beat the conventional methods of using Adaptive Filters. The analysis proves the superiority of the ANNs over the LMS filter by giving an accuracy of 95.72 % over 79% accuracy achieved by the LMS filter on the simulated data. And on practical data, the ANN kept its behaviour consistent by achieving 98.68% accuracy while LMS filter offered only 91% accuracy. Additionally, the neural network was also de-noising the signal while predicting. For the simulated data an SNR (Signal to Noise Ratio) of 29.71 dB and 16.33 dB for Neural Network prediction and LMS filter prediction were attained, respectively. Whereas, on real data, SNR values of 22.8 dB and 3.8 dB are delivered by the ANNs and LMS filter, respectively.

There will always be scope for improvement in areas such as time series forecasting. With the help of neural networks, much complex deep learning models can be applied to achieve exceptional accuracy and eventually make reliable forecasts. Proper implementation of the system may allow real-time prediction systems. Adopting this approach may even eradicate the need to de-noise the signal before analyzing. Constantly advancing computational technologies would complement biomedical applications with high compute capacity.

Out of all the possible applications for time series forecasting, it would be the most useful in biomedical applications where accuracy plays a prominent role. For example, the same forecasting can be made by training the ML model with real-time ECG data which will further allow us to track the recovery of a patient or abnormal behaviour of the heart maybe predicted well before, which would allow adequate time to take precautions. Hence, it would be apt to study the recovery of a patient with cardiac abnormalities and forecast the estimated date of complete recovery or predict a heart attack or stroke. Similar approach may even be used on other physiological data to improve the efficiency of healthcare systems.

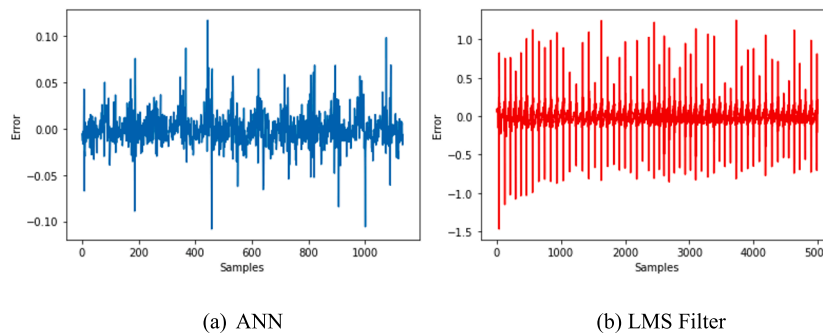


Fig. 14. Error vs sample graph.

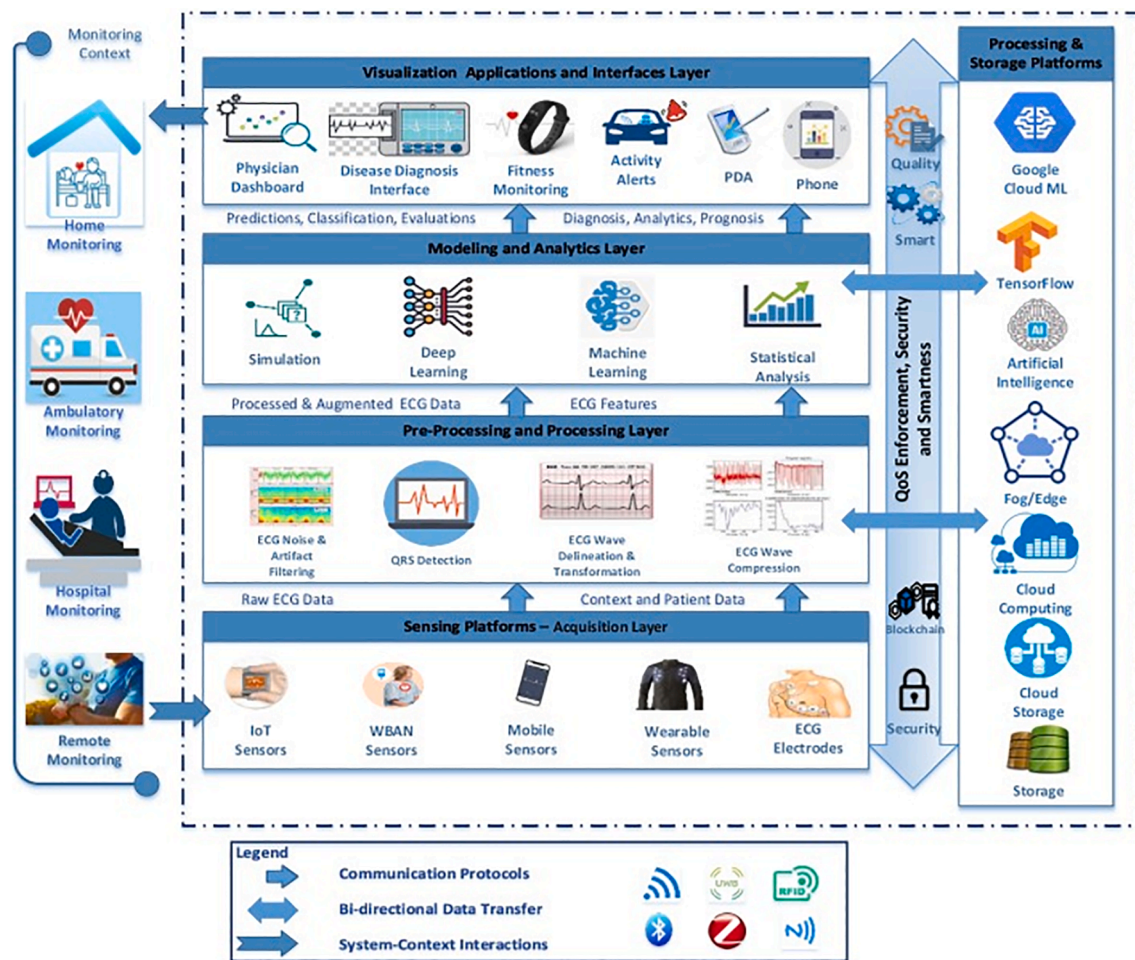


Fig. 15. Architecture of ECG monitoring systems.

CRediT authorship contribution statement

K. Ratna Prakarsha: Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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